

Instructions for *ACL Proceedings

Anonymous ACL submission

Abstract

This document is a supplement to the general instructions for *ACL authors. It contains instructions for using the L^AT_EX style files for ACL conferences. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. These instructions should be used both for papers submitted for review and for final versions of accepted papers.

1 Methodology

Our approach proceeds in two phases, illustrated in Figure ?? . **Phase 1** trains sparse autoencoders on the internal activations of frozen language models to decompose dense representations into interpretable sparse features. **Phase 2** uses those features at inference time to selectively remove target knowledge without modifying model weights.

1.1 Step 1: Models and Data

Models. We study GPT-2 Small (12 layers, $d_{\text{model}} = 768$, 117M parameters) and GPT-2 Medium (24 layers, $d_{\text{model}} = 1,024$, 345M parameters). These two models are chosen deliberately to represent contrasting regimes of representation structure. GPT-2 Small’s smaller capacity encourages relatively sparse, monosemantic representations, while GPT-2 Medium’s larger capacity allows it to exploit superposition encoding multiple concepts in shared feature directions. By studying both, we can ask whether SAE-based unlearning scales with model size and how the internal knowledge organisation affects selectivity.

TOFU benchmark. We evaluate on the TOFU benchmark (Maini et al., 2024), which contains question–answer pairs about 19 fictitious authors whose biographies were synthetically generated and injected into fine-tuned models. For each target author a , we define a forget set $\mathcal{F}_a$ (14–20 QA

pairs belonging to author a) and a retain set $\mathcal{R}_a$ (200 QA pairs sampled from the remaining 18 authors). Because the authors are entirely fictional, the knowledge boundaries are precise everything the model knows about a target author came from a controlled injection, making TOFU an ideal testbed for measuring selective forgetting.

1.2 Step 2: Activation Extraction

We run both models on the TOFU training set in frozen evaluation mode and collect residual stream activations at every transformer layer. Base model weights are never modified at any stage.

Different types of knowledge are localised at different depths in transformer models (Tenney et al., 2019). By extracting from all layers and training a separate SAE at each, we allow the layer-wise analysis to reveal where in the network author-specific knowledge is most concentrated rather than assuming a fixed target layer.

1.3 Step 3: SAE Training

We train a sparse autoencoder at every layer of both models across five dictionary sizes $D \in \{4,096, 8,192, 16,384, 32,768, 65,536\}$, yielding 60 SAEs for GPT-2 Small and 120 for GPT-2 Medium (180 total). Each SAE decomposes a d -dimensional activation vector $\mathbf{x}$ into a sparse code $\mathbf{z} \in \mathbb{R}^D$ and reconstruction $\hat{\mathbf{x}}$:

$$\mathbf{z} = \text{ReLU}(\mathbf{W}_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{dec}}) + \mathbf{b}_{\text{enc}}) \quad (1)$$

$$\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}} \mathbf{z} + \mathbf{b}_{\text{dec}} \quad (2)$$

trained to minimise $\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda \|\mathbf{z}\|_1$ with decoder columns constrained to unit norm.

Dense transformer activations are difficult to interpret because individual neurons are polysemantic (Bricken et al., 2023). The granularity of the SAE decomposition; i.e how finely it carves up activation space is controlled by D . A small D forces coarse features that may conflate distinct concepts;

a large D allows fine-grained features but risks fragmenting coherent concepts into redundant pieces. How a model responds to this granularity sweep is informative about its internal knowledge organisation, and directly predicts whether SAE-guided unlearning will be selective.

1.4 Step 4: SAE Validation

Each trained SAE is validated by computing the cosine similarity between original activations $\mathbf{x}$ and SAE reconstructions $\hat{\mathbf{x}}$. We require cosine similarity > 0.99 as the primary passing criterion. The validation is critical here because, if the SAE distorts the activation space, ablating features in the SAE’s basis would produce wrong results.

1.5 Step 5: Feature Scoring

For each target author (a) and each SAE feature (i), we compute a discriminative contrast score:

$$d_i = \bar{z}_i^{\mathcal{F}} - \bar{z}_i^{\mathcal{R}} \quad (3)$$

where $\bar{z}_i^{\mathcal{F}}$ and $\bar{z}_i^{\mathcal{R}}$ are the mean activations of feature i across all tokens in the forget and retain sets, respectively. Features with high d_i activate more strongly on the target author’s text than on the rest of the corpus. We need discriminative features, those are active for the target but not for the retain set and the contrast score $d_i = \bar{z}_i^{\mathcal{F}} - \bar{z}_i^{\mathcal{R}}$ directly measures this specificity.

1.6 Step 6: Feature Selection

We select the top- k features with the highest discriminative scores for ablation, evaluated at $k \in \{50, 100, 200, 500\}$. This analysis reveals how author-specific knowledge is distributed in the model’s representation. If knowledge is concentrated in a small number of features (sparse organisation), unlearning will plateau at small k . If knowledge is distributed across many features (distributed organisation), selectivity will scale with k but collateral damage will also grow. Therefore this analysis provides a behavioural signature of the model’s internal knowledge structure, complementing the dictionary-size analysis.

1.7 Step 7: Unlearning using SAE

During inference, we register a forward hook at a target layer (ℓ) that intercepts the model’s hidden state ($\mathbf{h}$), routes it through the SAE, modifies the selected features, and reconstructs the activation. The modified activation replaces the original in the model’s forward pass. We evaluate

three ablation strategies for the selected feature set $\mathcal{S} = \{i_1, \dots, i_k\}$:

- **Zero ablation:** Selected features are clamped to zero ($z'_i = 0$ for $i \in \mathcal{S}$), completely removing the contribution of those feature directions from the representation.

1.8 Causal Ablation Study

To establish a causal link between SAE features and author-specific knowledge, we evaluate three ablation conditions at each value of $k \in \{50, 100, 200, 500\}$ using SAEs trained at $D = 65536$:

1. **Forward forget ablation:** Top- k forget-discriminative features are zero-ablated during inference on the forget set. A successful intervention reduces ROUGE-L relative to baseline.
2. **Forward retain ablation:** The same top- k forget-discriminative features are zero-ablated during inference on the retain set. A selective intervention should leave retain performance relatively intact.
3. **Reverse forget ablation:** Top- k retain-discriminative features are zero-ablated during inference on the forget set. This control tests whether the discriminative scoring identifies features with genuine author-specific causal roles.

We report ROUGE-L between model generations and ground-truth answers, averaged across all 19 authors.

1.9 Evaluation Metrics

We evaluate unlearning quality using two metrics, applied at different stages of analysis.

ROUGE-L. We compute ROUGE-L (Lin, 2004) between model generations and ground-truth answers on both the forget set $\mathcal{F}_a$ and retain set $\mathcal{R}_a$. Effective unlearning reduces forget-set ROUGE-L (the model can no longer reproduce target knowledge) while preserving retain-set ROUGE-L (general knowledge remains intact). ROUGE-L is used as the primary metric for the causal ablation study.

Perplexity-based selectivity. For the layer-wise and dictionary-size analyses, where we evaluate across all 180 SAE configurations, we define selectivity using perplexity:

$$S = \frac{\Delta_{\text{forget}}}{|\Delta_{\text{retain}}| + \epsilon} = \frac{\text{PPL}_{\text{forget}}^{\text{ablated}} - \text{PPL}_{\text{forget}}^{\text{base}}}{|\text{PPL}_{\text{retain}}^{\text{ablated}} - \text{PPL}_{\text{retain}}^{\text{base}}| + \epsilon} \quad (4)$$

where $\epsilon = 10^{-8}$ prevents division by zero when retain perplexity is unchanged. The numerator measures how much harder the model finds the forget set after ablation (higher perplexity = less knowledge retained). The denominator measures collateral damage to the retain set. $S > 1$ indicates the intervention disproportionately targets the intended knowledge; $S = 1$ corresponds to indiscriminate degradation; $S < 1$ indicates greater collateral damage than forgetting.

ROUGE-L is computed from model generations and directly measures whether the model can reproduce the target content the most faithful measure of unlearning quality. However, generating text for all 19 authors across 180 SAE configurations and multiple values of k is computationally prohibitive. Perplexity is computed from a single forward pass and scales to the full configuration sweep, enabling the layer-wise and dictionary-size analyses. Both metrics are consistent in direction: configurations with high ROUGE-L selectivity also achieve high perplexity-based selectivity, confirming that the two measures capture the same underlying phenomenon.

1.10 Step 9: Circuit Analysis

In parallel with the ablation evaluation, we conduct a circuit analysis to identify which attention heads mediate the forget-discriminative SAE features. For the top-10 forget-discriminative features at $D = 65,536$ and the primary analysis layer (layer 10 for Small, layer 23 for Medium), we compute the Pearson correlation between each attention head’s output norm and each feature’s activation, per token, across all tokens in the forget set:

$$r_{hf} = \text{Pearson}(\|\text{head}_h(\mathbf{x})\|_2, z_f(\mathbf{x})) \quad (5)$$

for head (h) and feature (f). This yields a correlation matrix of shape ($n_{\text{heads}} \times n_{\text{features}}$) per author. We average r_{hf} across features to obtain a per-head importance score, and classify heads as *promoters* ($r > 0$, consistently amplify author-specific features) or *suppressors* ($r < 0$, consistently attenuate them) based on sign polarity across

all five evaluated authors. We define strongly correlated heads as those with mean $|\bar{r}| > 0.3$.

The ablation experiment establishes that ablating top- k features selectively removes author-specific knowledge (causal evidence at the feature level). Circuit analysis goes one level deeper: it asks *how* these features acquire their values which computational components write the information into the SAE features. Identifying the attention heads that mediate the discriminative features provides three benefits:

1. **Mechanistic grounding.** If the same set of attention heads mediates discriminative features across all 19 authors (author-invariant circuit), this confirms that the model uses a shared computational pathway for encoding author-specific knowledge, rather than a bespoke mechanism per author.
2. **Validation of feature scoring.** If the circuit is concentrated in a small number of heads (rather than uniformly distributed), this corroborates that the discriminative scoring has identified genuine knowledge-encoding features, not generic activation patterns.
3. **Explanation of unlearning behaviour.** The promoter-suppressor structure explains *why* ablating features at the target layer is effective: by that depth, the circuit’s contributions have been integrated into a small number of identifiable feature directions that the SAE can access and zero out.

2 Results

2.1 SAE Training

2.2 SAE Based Unlearning

We evaluate SAE-guided unlearning across all layers and all 19 TOFU authors for each dictionary size, computing perplexity based selectivity ($S = \Delta_{\text{forget}}/\Delta_{\text{retain}}$) per layer and averaging across layers and authors to obtain a single summary statistic per model dictionary configuration.

Table 2 and Figure ??(a) reveal that the two models respond to increasing dictionary size in opposite directions:

- **GPT-2 Small** exhibits a monotonic decrease in selectivity, from 2.33 at $D = 4096$ to 2.17 at $D = 65536$.

Table 1: SAE training quality across all configurations. **Cos**: cosine similarity between original and reconstructed activations (mean across layers; minimum in parentheses). All 180 SAEs satisfy the primary passing criterion of $\cos > 0.99$.

Model	D	Cos (min accross layers)	Pass rate
GPT-2 Small	4K	0.9998 (0.9991)	12/12
	8K	0.9996 (0.9980)	12/12
	16K	0.9992 (0.9969)	12/12
	32K	0.9980 (0.9930)	12/12
	64K	0.9982 (0.9957)	12/12
GPT-2 Medium	4K	0.9999 (0.9993)	24/24
	8K	0.9998 (0.9993)	24/24
	16K	0.9997 (0.9992)	24/24
	32K	0.9994 (0.9977)	24/24
	64K	0.9989 (0.9977)	24/24
GPT-2 Large	4K	0.9995 (0.9991)	36/36
	8K	0.9994 (0.9990)	36/36
	16K	0.9996 (0.9992)	36/36
	32K	0.9994 (0.9972)	36/36
	64K	0.9989 (0.9972)	36/36

Table 2: Mean selectivity (averaged across all layers and 19 authors) as a function of dictionary size. GPT-2 Small’s selectivity decreases monotonically while GPT-2 Medium’s increases, revealing opposite responses to finer-grained decomposition.

D	GPT-2		
	Medium	Small	Large
4K	2.55	2.33	1.08
8K	2.55	2.30	1.09
16K	2.62	2.25	4.31
32K	2.92	2.20	5.17
64K	3.48	2.17	6.22
Δ (4K $\rightarrow$ 64K)	+0.93	-0.16	+5.14

- **GPT-2 Medium** exhibits a monotonic increase, from 2.55 at $D = 4096$ to 3.48 at $D = 65536$.

Both models remain above the indiscriminate baseline of $S = 1$ at all dictionary sizes, confirming that SAE-guided ablation achieves selective unlearning across all configurations. However, the *direction* of the scaling trend is reversed: larger dictionaries help Medium but hurt Small.

Quantifying the asymmetry: Medium’s selectivity gain from $D = 4096$ to $D = 65536$ is +0.93, while Small’s loss is -0.16. The improvement from resolving superposition in the larger model far exceeds the degradation from fragmenting already-clean features in the smaller model. This asymme-

try holds at every step of the dictionary-size sweep: Medium’s selectivity is flat through $D = 8192$, then accelerates sharply between $D = 16384$ and $D = 65536$, suggesting a capacity threshold below which the SAE cannot exploit Medium’s superposed structure. Small’s decline, by contrast, is gradual and roughly linear on a log scale, consistent with incremental fragmentation of already-separated features.

2.3 Baseline: Parameter-Space Unlearning

We evaluate three parameter-space unlearning methods that represent distinct optimization strategies, each run across 9 hyperparameter configurations (learning rates $\in \{10^{-5}, 5 \times 10^{-5}, 10^{-4}\}$, steps $\in \{5, 10, 20\}$) for all 19 TOFU authors on both GPT-2 Small and GPT-2 Medium, yielding 513 runs per model.

Gradient Ascent (GA) (Jang et al., 2023): Maximizes the loss on the forget set to push the model away from generating memorized content: $\mathcal{L} = -\mathcal{L}_{\text{CE}}(\mathcal{F})$.

Gradient Difference (GradDiff) (Maini et al., 2024): Combines gradient ascent on the forget set with gradient descent on the retain set, explicitly optimizing for both forgetting and preservation: $\mathcal{L} = -\mathcal{L}_{\text{CE}}(\mathcal{F}) + \beta \cdot \mathcal{L}_{\text{CE}}(\mathcal{R})$, with $\beta = 1.0$.

KL Minimization (IDK) (Maini et al., 2024): Pushes forget-set outputs toward an uninformative response (“I don’t know”) while anchoring retain-set behavior to the original model via KL divergence: $\mathcal{L} = \mathcal{L}_{\text{CE}}(\mathcal{F}, \text{“IDK”}) + \beta \cdot D_{\text{KL}}(p_{\theta'} \| p_{\theta} | \mathcal{R})$, with $\beta = 1.0$.

All three methods use Adam with gradient clipping at norm 1.0.

Results: Table 3 summarizes the outcomes. Each method fails on GPT-2 Medium, but in a qualitatively different way and none achieves the controlled, selective forgetting that SAE-guided ablation provides.

Gradient Ascent destroys both models. On GPT-2 Small, 60.8% of configurations produce catastrophic retain-set degradation ($\text{PPL} > 10^4$). On GPT-2 Medium, this rises to 77.8%, and no configuration keeps retain PPL below $2 \times$ baseline while achieving meaningful forgetting. At higher learning rates ($\geq 5 \times 10^{-5}$), both forget and retain PPL explode to $\sim 10^{15}$, indicating complete model collapse. The only non-catastrophic configurations (lr

$= 10^{-5}$, steps ≤ 10) produce selectivities of 1.3–1.5 on Small and 1.5–2.1 on Medium well below SAE-guided ablation.

Gradient Difference avoids catastrophic retain collapse (only 0.6% of runs on Small, 1.8% on Medium exceed retain PPL $> 10^4$), but produces pathological forgetting. Forget PPL routinely reaches 10^3 – 10^{15} , meaning the model becomes incoherent on the forget set rather than selectively forgetting. The numerically high selectivity values (up to 10^{15}) are misleading: they arise because retain PPL barely changes while forget PPL explodes, but a model that generates gibberish on the forget set has not “unlearned” in any meaningful sense it has simply broken. The enormous variance across authors (standard deviations exceeding the mean by orders of magnitude) further indicates that the intervention is unstable and unpredictable.

KL Minimization is the most conservative method: zero catastrophic failures on either model. However, it achieves almost no forgetting. On GPT-2 Small, forget PPL decreases in most configurations (mean $\Delta = -1.24$), indicating the model is not losing knowledge at all. On GPT-2 Medium, the best viable configuration achieves a forget PPL increase of only 1.3 (from 25.5 to 28.1) with a selectivity of 1.3. The KL anchor on the retain set is so strong that the model effectively resists any change.

2.4 Causal Ablation

To confirm that the selectivity divergence reflects genuine differences in knowledge organization not artifacts of the SAE or the scoring heuristic we conduct the three-condition causal ablation study described in 1 at $D = 65536$. Table 4 reports the results.

GPT-2 Small saturates at $k = 50$: Forward forget ROUGE-L is effectively constant across all values of k (≈ 0.204), as is forward retain ROUGE-L (≈ 0.203). Selectivity holds steady at ≈ 1.53 from $k = 50$ through $k = 500$. This demonstrates that the top-50 forget-discriminative features capture the entirety of Small’s removable author-specific knowledge; additional features are causally inert.

GPT-2 Medium degrades steadily with k : Forward forget ROUGE-L drops from 0.189 at $k = 50$ to 0.129 at $k = 500$, indicating that author knowledge is distributed across a large set of causally active features. However, retain ROUGE-L degrades in tandem ($0.213 \rightarrow 0.154$), eroding selectivity

from 3.26 to 1.60. The larger model benefits from a higher selectivity ceiling its top-50 features are more precisely targeted than Small’s but this advantage diminishes as more features are ablated.

Reverse control validates discriminative scoring: For both models, reverse forget ROUGE-L (ablating retain-discriminative features on the forget set) is consistently higher than forward forget ROUGE-L (ablating forget-discriminative features on the forget set). This confirms that the discriminative score identifies features with genuine author-specific causal roles: ablating features associated with other authors has less impact on the target author’s outputs than ablating the target author’s own features.

2.5 Unified Interpretation: Sparse vs. Distributed Knowledge

The dictionary-size and k -axis results jointly reveal the same underlying property representation sparsity viewed from two complementary directions. Table 2 summarizes the contrasting behaviors.

Feature fragmentation (Small): GPT-2 Small’s representations are already relatively monosemantic: individual features correspond to coherent concepts. Expanding the dictionary does not resolve superposition it fragments already-coherent features into redundant pieces. Knowledge is spread across more features without improving precision, diluting selectivity. The k -axis confirms this: ablation saturates at $k = 50$ because the sparse knowledge budget is exhausted; additional features are causally inert copies.

Superposition resolution (Medium): GPT-2 Medium packs more concepts into shared feature directions. Larger dictionaries give the SAE enough capacity to disentangle these superposed representations into distinct, cleanly separable features. Each author’s knowledge becomes more isolated, improving selectivity. But the k -axis reveals the cost: because knowledge is distributed across more causally active features, ablating more of them inevitably incurs collateral damage, and selectivity decays from 3.26 to 1.60.

Asymmetric effect magnitude: Medium’s selectivity gain across dictionary sizes (+0.88) is $2.6\times$ the magnitude of Small’s loss (-0.30). This asymmetry follows from the nature of each mechanism: superposition resolution disentangles actively *entangled* concepts, yielding a large selectivity gain;

Table 3: Parameter-space baseline results on GPT-2 Small and Medium, across 9 hyperparameter configurations $\times$ 19 authors per method (171 runs each). **Catastrophic**: % of runs where retain PPL exceeds 10^4 . **Best viable**: highest mean selectivity among configurations where retain PPL stays below $2\times$ baseline and forget PPL stays below 10^4 (i.e., the model remains coherent). SAE selectivity is shown for comparison.

Model	Method	All Configurations (171 runs)				Best Viable Config			
		Catastrophic	Med. Δ_F	Med. Δ_R	Med. Sel.	Config	Sel.	Failure Mode	SAE Sel.
GPT-2 Small	GA	60.8%	6.2×10^8	8.9×10^7	1.48	$\text{lr}=10^{-5}, T=5$	1.18	Model collapse	
	GradDiff	0.6%	2.4×10^3	4.4	174.6	$\text{lr}=10^{-5}, T=20$	178*	Incoherent forget	2.17–2.33
	KL/IDK	0.0%	−1.5	−1.4	−0.9	–	≤ 1.04	No forgetting	
GPT-2 Medium	GA	77.8%	1.6×10^{15}	1.6×10^{15}	1.00	–	None viable	Total collapse	
	GradDiff	1.8%	1.5×10^5	13.0	3.5×10^3	$\text{lr}=10^{-5}, T=5$	61*	Incoherent forget	2.55–3.48
	KL/IDK	0.0%	1.4	1.5	0.77	$\text{lr}=10^{-4}, T=10$	1.34	Minimal forgetting	

*GradDiff selectivity is inflated: forget PPL reaches 10^2 – 10^3 (incoherent generation) while retain PPL barely changes. These values do not represent meaningful selective unlearning.

Table 4: Causal ablation at $D = 65536$, averaged across 19 authors. **Fwd. F / Fwd. R**: ROUGE-L after ablating top- k forget-discriminative features on forget / retain sets. **Rev. F**: ROUGE-L after ablating top- k retain-discriminative features on the forget set. Selectivity = $\Delta_{\text{forget}} / \Delta_{\text{retain}}$.

Model	k	Fwd. F	Δ	Fwd. R	Δ	Rev. F	Δ	Sel.
<i>Baselines (no ablation)</i>								
Small	–	0.251	–	0.234	–	0.251	–	–
Medium	–	0.257	–	0.234	–	0.257	–	–
Small	50	0.204	−0.047	0.203	−0.031	0.219	−0.032	1.54
	100	0.204	−0.047	0.203	−0.031	0.219	−0.032	1.53
	200	0.205	−0.047	0.203	−0.031	0.219	−0.032	1.52
	500	0.205	−0.047	0.203	−0.031	0.219	−0.032	1.51
Medium	50	0.189	−0.068	0.213	−0.021	0.208	−0.049	3.26
	100	0.185	−0.073	0.196	−0.037	0.200	−0.057	1.94
	200	0.167	−0.090	0.183	−0.051	0.189	−0.068	1.76
	500	0.129	−0.128	0.154	−0.080	0.169	−0.088	1.60

fragmentation merely adds redundancy to already-separated concepts, producing gradual degradation.

Convergence at small dictionaries: At $D = 4096$, both models achieve similar selectivity (≈ 2.5). The dictionary is the bottleneck: neither model’s SAE has enough capacity to exploit structural differences, forcing both into similarly coarse decompositions. The divergence emerges only when dictionary capacity is sufficient for model-specific structure to matter.

Connection to gradient ascent failure: The distributed, superposed nature of GPT-2 Medium’s knowledge organization explains why gradient ascent fails catastrophically on this model. Gradient updates propagate through all parameters simultaneously and cannot be confined to the subset of directions encoding target knowledge when that knowledge is entangled with unrelated concepts. SAE-guided ablation succeeds precisely because it operates in the *disentangled* feature space rather than the raw parameter space, enabling surgical

removal of specific knowledge directions identified by the discriminative scoring.

2.6 Circuit Analysis: Attention Heads Mediating Author-Specific Features

The causal ablation study (2.4) establishes that SAE features are causally responsible for author-specific outputs, and the layer-wise analysis [?] identifies where in the network these features are most discriminative. A natural follow-up question is how these features participate in the model’s computation: which attention heads write the information that becomes the forget-discriminative features at the target layer?

Method: For each of the top-10 forget-discriminative features (at $D = 65536$, layer 10) for 5 authors, we compute the Pearson correlation between each attention head’s output norm (per token) and the feature’s activation (per token) across all tokens in the author’s forget set. This yields a correlation matrix of shape (132 heads $\times$ 10 features) per author, revealing which heads’ outputs co-vary most strongly with the discriminative features.

A shared, author-invariant circuit: The head-feature correlations are remarkably consistent across authors. Kendall rank correlation between all pairs of authors’ head importance rankings ranges from $\tau = 0.59$ to $\tau = 0.75$ (all $p < 10^{-23}$), indicating that the same heads mediate author-specific features regardless of which author is being targeted. This rules out the possibility that the circuit is idiosyncratic to individual authors; instead, the model uses a **shared architectural pathway** for encoding author-specific knowledge.

Table 5: Top 8 attention heads correlated with forget-discriminative SAE features in GPT-2 Small. Mean $|\bar{r}|$ is averaged across 10 features $\times$ 5 authors. **Sign**: polarity of the correlation, consistent across all 5 authors. **Role**: promoter heads amplify author-specific features; suppressor heads attenuate them.

Head	Layer	Mean $ \bar{r} $	Sign	Role
L1.H3	1	0.540	−	Suppressor
L2.H7	2	0.500	−	Suppressor
L1.H9	1	0.409	−	Suppressor
L2.H9	2	0.373	−	Suppressor
L1.H2	1	0.373	+	Promoter
L4.H4	4	0.353	−	Suppressor
L3.H6	3	0.350	−	Suppressor
L6.H2	6	0.350	+	Promoter

Concentration in a small set of heads: Of the 132 attention heads in GPT-2 Small (12 layers $\times$ 12 heads), only 13 exhibit strong mean correlations ($|\bar{r}| > 0.3$), while 89 are weak ($|\bar{r}| \leq 0.2$). The top 8 heads achieve a mean $|\bar{r}|$ of 0.41, $2.7\times$ the median head and $7.1\times$ the bottom 8 heads indicating that the causal influence of forget-discriminative features is mediated by a concentrated subset of the model’s attention mechanism.

Promoter-suppressor architecture: The top 8 heads maintain perfectly consistent sign polarity across all 5 authors (Table 5): two heads (L1.H2, L6.H2) are uniformly *positively* correlated with discriminative features (promoters), while six heads (L1.H3, L2.H7, L1.H9, L2.H9, L4.H4, L3.H6) are uniformly *negatively* correlated (suppressors). No head switches sign across authors.

This suggests an antagonistic circuit: promoter heads write author-specific information into the residual stream, while suppressor heads attenuate it. The balance between these opposing contributions at layer 10 determines how strongly the model expresses author-specific knowledge and, consequently, how effectively SAE-guided ablation can target it.

Layer distribution: The circuit is concentrated in the early layers of the network. Of the top 20 heads by mean $|\bar{r}|$, 5 are in layer 1, 3 in layer 2, and 3 in layer 9 with layer 1 alone contributing more strongly correlated heads than any other layer. This suggests a two-stage architecture: early-layer heads (layers 1-4) perform the primary encoding of author-specific information, while late-layer heads

(layer 9) provide a secondary refinement just before the target SAE layer (layer 10), where the discriminative features are most localized.

Connection to unlearning: The promoter-suppressor structure has a direct implication for SAE-based unlearning. When we zero-ablate forget-discriminative features at layer 10, we are removing the *net output* of this antagonistic circuit. The fact that the circuit is shared across authors explains why our contrast scoring (1.7) succeeds: the scoring does not need to discover a separate mechanism per author because the underlying circuit is the same only the specific features it writes to differ. The concentration of the circuit in early layers further explains why ablation at layer 10 is effective: by this depth, the promoter and suppressor contributions have been integrated into a small number of feature directions that are accessible to the SAE.

Acknowledgments

This document has been adapted by Steven Bethard, Ryan Cotterell and Rui Yan from the instructions for earlier ACL and NAACL proceedings, including those for ACL 2019 by Douwe Kiela and Ivan Vulić, NAACL 2019 by Stephanie Lukin and Alla Roskovskaya, ACL 2018 by Shay Cohen, Kevin Gimpel, and Wei Lu, NAACL 2018 by Margaret Mitchell and Stephanie Lukin, BibTeX suggestions for (NA)ACL 2017/2018 from Jason Eisner, ACL 2017 by Dan Gildea and Min-Yen Kan, NAACL 2017 by Margaret Mitchell, ACL 2012 by Maggie Li and Michael White, ACL 2010 by Jing-Shin Chang and Philipp Koehn, ACL 2008 by Johanna D. Moore, Simone Teufel, James Allan, and Sadaoki Furui, ACL 2005 by Hwee Tou Ng and Kemal Oflazer, ACL 2002 by Eugene Charniak and Dekang Lin, and earlier ACL and EACL formats written by several people, including John Chen, Henry S. Thompson and Donald Walker. Additional elements were taken from the formatting instructions of the *International Joint Conference on Artificial Intelligence* and the *Conference on Computer Vision and Pattern Recognition*.

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A Example Appendix

This is an appendix.